Newman-Penrose Formalism Calculations

Stephani et al., Equation (35.30_e-1)

Metric

$$
\begin{aligned}
ds^2 = \Bigg(& -\frac{2\left(K^4+4\,K^3+6\,K^2+4\,K+1\right)v^2\,x^4}{\mathcal{D}_1} + \frac{2\left(lK^2+2\,lK+l\right)v\,x^4}{\mathcal{D}_1} + \frac{2\left(K^2+2\,K+1\right)H\,x^4}{\mathcal{D}_1} - \frac{2\left(K^4+4\,K^3+6\,K^2+4\,K+1\right)v^2\,y^4}{\mathcal{D}_1} \\
& + \frac{2\left(lK^2+2\,lK+l\right)v\,y^4}{\mathcal{D}_1} + \frac{2\left(K^2+2\,K+1\right)H\,y^4}{\mathcal{D}_1} - \frac{2\left(K^4+4\,K^3+6\,K^2+4\,K+1\right)v^2}{\mathcal{D}_1} + \frac{4\left(K^4+4\,K^3+6\,K^2+4\,K+1\right)v^2\,x^2}{\mathcal{D}_1} - \frac{4\left(lK^2+2\,lK+l\right)v\,x^2}{\mathcal{D}_1} \\
& - \frac{4\left(K^2+2\,K+1\right)H\,x^2}{\mathcal{D}_1} + \frac{4\left(K^4+4\,K^3+6\,K^2+4\,K+1\right)v^2\,y^2}{\mathcal{D}_1} - \frac{4\left(\left(K^4+4\,K^3+6\,K^2+4\,K+1\right)v^2 - \left(lK^2+2\,lK+l\right)v - \left(K^2+2\,K+1\right)H\right)x^2y^2}{\mathcal{D}_1} \\
& - \frac{4\left(lK^2+2\,lK+l\right)v\,y^2}{\mathcal{D}_1} - \frac{4\left(K^2+2\,K+1\right)H\,y^2}{\mathcal{D}_1} + \frac{2\left(lK^2+2\,lK+l\right)v}{\mathcal{D}_1} + \frac{2\left(K^2+2\,K+1\right)H}{\mathcal{D}_1} - \frac{2\,b_x\left(u,x,y\right)^2}{\mathcal{D}_1} - \frac{2\,b_y\left(u,x,y\right)^2}{\mathcal{D}_1}\Bigg)\mathrm{d}u\otimes\mathrm{d}u - \mathrm{d}u \\
& \otimes\mathrm{d}v - \frac{2\,b_x\left(u,x,y\right)}{\mathcal{D}_2}\,\mathrm{d}u\otimes\mathrm{d}x - \frac{2\,b_y\left(u,x,y\right)}{\mathcal{D}_2}\,\mathrm{d}u\otimes\mathrm{d}y - \mathrm{d}v\otimes\mathrm{d}u - \frac{2\,b_x\left(u,x,y\right)}{\mathcal{D}_2}\,\mathrm{d}x\otimes\mathrm{d}u - \frac{2}{\mathcal{D}_2}\,\mathrm{d}x\otimes\mathrm{d}x - \frac{2\,b_y\left(u,x,y\right)}{\mathcal{D}_2}\,\mathrm{d}y\otimes\mathrm{d}u - \frac{2}{\mathcal{D}_2}\,\mathrm{d}y\otimes\mathrm{d}y
\end{aligned}
$$

$$\mathcal{D}_1 = \left(K^2+2\,K+1\right)x^4 + \left(K^2+2\,K+1\right)y^4 - 2\left(K^2+2\,K+1\right)x^2 + 2\left(\left(K^2+2\,K+1\right)x^2 - K^2 - 2\,K - 1\right)y^2 + K^2 + 2\,K + 1,$$

$$\mathcal{D}_2 = \left(K^2+2\,K+1\right)x^2 + \left(K^2+2\,K+1\right)y^2 - K^2 - 2\,K - 1.$$

Coordinates

$(x^\mu) = (u, v, x, y)$

Metric Functions

$K(u) = K$

$H(u, x, y) = H$

Null Tetrad

Covariant components

$l = (1)\ du$

$$n = \left(v^2K^2 + 2\,v^2K - lv + v^2 + \frac{b_x\,(u,x,y)^2}{(x^4 + 2\,x^2y^2 + y^4 - 2\,x^2 - 2\,y^2 + 1)(K^2 + 2\,K + 1)} + \frac{b_y\,(u,x,y)^2}{(x^4 + 2\,x^2y^2 + y^4 - 2\,x^2 - 2\,y^2 + 1)(K^2 + 2\,K + 1)} - H\right)\ du$$
$$+ (1)\ dv + \left(\frac{2\,b_x\,(u,x,y)}{(x^2 + y^2 - 1)(K^2 + 2\,K + 1)}\right)\ dx + \left(\frac{2\,b_y\,(u,x,y)}{(x^2 + y^2 - 1)(K^2 + 2\,K + 1)}\right)\ dy$$

$$m = \left(\frac{\sqrt{-\frac{1}{x^2+y^2-1}}}{K+1}\right)\ dx + \left(\frac{i\,\sqrt{-\frac{1}{x^2+y^2-1}}}{K+1}\right)\ dy$$

$$\overline{m} = \left(\frac{\sqrt{-\frac{1}{x^2+y^2-1}}}{K+1}\right)\ dx + \left(-\frac{i\,\sqrt{-\frac{1}{x^2+y^2-1}}}{K+1}\right)\ dy$$

Contravariant components

$l = (-1)\ dv$

$$n = -\,\mathrm{d}u + \frac{v^2x^4K^4}{\mathcal{D}_1} + \frac{2\,v^2x^2y^2K^4}{\mathcal{D}_1} + \frac{v^2y^4K^4}{\mathcal{D}_1} + \frac{4\,v^2x^4K^3}{\mathcal{D}_1} + \frac{8\,v^2x^2y^2K^3}{\mathcal{D}_1} + \frac{4\,v^2y^4K^3}{\mathcal{D}_1} - \frac{lvx^4K^2}{\mathcal{D}_1} + \frac{6\,v^2x^4K^2}{\mathcal{D}_1} - \frac{2\,lvx^2y^2K^2}{\mathcal{D}_1}$$
$$+ \frac{12\,v^2x^2y^2K^2}{\mathcal{D}_1} - \frac{lvy^4K^2}{\mathcal{D}_1} + \frac{6\,v^2y^4K^2}{\mathcal{D}_1} - \frac{2\,v^2x^2K^4}{\mathcal{D}_1} - \frac{2\,v^2y^2K^4}{\mathcal{D}_1} - \frac{2\,lvx^4K}{\mathcal{D}_1} + \frac{4\,v^2x^4K}{\mathcal{D}_1} - \frac{4\,lvx^2y^2K}{\mathcal{D}_1} + \ldots$$

$$m = \left(\frac{b_x\,(u,x,y)}{\sqrt{|x^2 + y^2 - 1|}|K+1|} + \frac{i\,b_y\,(u,x,y)}{\sqrt{|x^2 + y^2 - 1|}|K+1|}\right)\ dv + \left(-\frac{x^2K^2}{2\,\sqrt{|x^2 + y^2 - 1|}|K+1|} - \frac{y^2K^2}{2\,\sqrt{|x^2 + y^2 - 1|}|K+1|} - \frac{x^2K}{\sqrt{|x^2 + y^2 - 1|}|K+1|} - \frac{y^2K}{\sqrt{|x^2 + y^2 - 1|}|K+1|}\right.$$
$$\left. - \frac{x^2}{2\,\sqrt{|x^2 + y^2 - 1|}|K+1|} - \frac{y^2}{2\,\sqrt{|x^2 + y^2 - 1|}|K+1|} + \frac{K^2}{2\,\sqrt{|x^2 + y^2 - 1|}|K+1|} + \frac{K}{\sqrt{|x^2 + y^2 - 1|}|K+1|} + \frac{1}{2\,\sqrt{|x^2 + y^2 - 1|}|K+1|}\right)\ dx$$
$$+ \left(-\frac{i\,x^2K^2}{2\,\sqrt{|x^2 + y^2 - 1|}|K+1|} - \frac{i\,y^2K^2}{2\,\sqrt{|x^2 + y^2 - 1|}|K+1|} - \frac{i\,x^2K}{\sqrt{|x^2 + y^2 - 1|}|K+1|} - \frac{i\,y^2K}{\sqrt{|x^2 + y^2 - 1|}|K+1|} - \frac{i\,x^2}{2\,\sqrt{|x^2 + y^2 - 1|}|K+1|} - \frac{i\,y^2}{2\,\sqrt{|x^2 + y^2 - 1|}|K+1|}\right.$$
$$\left. + \frac{i\,K^2}{2\,\sqrt{|x^2 + y^2 - 1|}|K+1|} + \frac{i\,K}{\sqrt{|x^2 + y^2 - 1|}|K+1|} + \frac{i}{2\,\sqrt{|x^2 + y^2 - 1|}|K+1|}\right)\ dy$$

$$\overline{m} = \left(\frac{b_x(u,x,y)}{\sqrt{|x^2+y^2-1|}|K+1|} - \frac{i\,b_y(u,x,y)}{\sqrt{|x^2+y^2-1|}|K+1|}\right) dv + \left(-\frac{x^2K^2}{2\sqrt{|x^2+y^2-1|}|K+1|} - \frac{y^2K^2}{2\sqrt{|x^2+y^2-1|}|K+1|} - \frac{x^2K}{\sqrt{|x^2+y^2-1|}|K+1|} - \frac{y^2K}{\sqrt{|x^2+y^2-1|}|K+1|}\right.$$
$$\left. - \frac{x^2}{2\sqrt{|x^2+y^2-1|}|K+1|} - \frac{y^2}{2\sqrt{|x^2+y^2-1|}|K+1|} + \frac{K^2}{2\sqrt{|x^2+y^2-1|}|K+1|} + \frac{K}{\sqrt{|x^2+y^2-1|}|K+1|} + \frac{1}{2\sqrt{|x^2+y^2-1|}|K+1|}\right) dx$$
$$+ \left(\frac{i\,x^2K^2}{2\sqrt{|x^2+y^2-1|}|K+1|} + \frac{i\,y^2K^2}{2\sqrt{|x^2+y^2-1|}|K+1|} + \frac{i\,x^2K}{\sqrt{|x^2+y^2-1|}|K+1|} + \frac{i\,y^2K}{\sqrt{|x^2+y^2-1|}|K+1|} + \frac{i\,x^2}{2\sqrt{|x^2+y^2-1|}|K+1|} + \frac{i\,y^2}{2\sqrt{|x^2+y^2-1|}|K+1|}\right.$$
$$\left. - \frac{i\,K^2}{2\sqrt{|x^2+y^2-1|}|K+1|} - \frac{i\,K}{\sqrt{|x^2+y^2-1|}|K+1|} - \frac{i}{2\sqrt{|x^2+y^2-1|}|K+1|}\right) dy$$

Directional Derivatives

$D = -\partial_v$

$$
\begin{aligned}
\Delta = -\partial_u &+ \frac{v^2 x^4 K^4}{x^4K^2 + 2\,x^2y^2K^2 + y^4K^2 + 2\,x^4K + 4\,x^2y^2K + 2\,y^4K + x^4 + 2\,x^2y^2 + y^4 - 2\,x^2K^2 - 2\,y^2K^2 - 4\,x^2K - 4\,y^2K - 2\,x^2 - 2\,y^2 + K^2 + 2\,K + 1} \\
&+ \frac{2\,v^2x^2y^2K^4}{x^4K^2 + 2\,x^2y^2K^2 + y^4K^2 + 2\,x^4K + 4\,x^2y^2K + 2\,y^4K + x^4 + 2\,x^2y^2 + y^4 - 2\,x^2K^2 - 2\,y^2K^2 - 4\,x^2K - 4\,y^2K - 2\,x^2 - 2\,y^2 + K^2 + 2\,K + 1} \\
&+ \frac{v^2y^4K^4}{x^4K^2 + 2\,x^2y^2K^2 + y^4K^2 + 2\,x^4K + 4\,x^2y^2K + 2\,y^4K + x^4 + 2\,x^2y^2 + y^4 - 2\,x^2K^2 - 2\,y^2K^2 - 4\,x^2K - 4\,y^2K - 2\,x^2 - 2\,y^2 + K^2 + 2\,K + 1} \\
&+ \frac{4\,v^2x^4K^3}{x^4K^2 + 2\,x^2y^2K^2 + y^4K^2 + 2\,x^4K + 4\,x^2y^2K + 2\,y^4K + x^4 + 2\,x^2y^2 + y^4 - 2\,x^2K^2 - 2\,y^2K^2 - 4\,x^2K - 4\,y^2K - 2\,x^2 - 2\,y^2 + K^2 + 2\,K + 1} \\
&+ \frac{8\,v^2x^2y^2K^3}{x^4K^2 + 2\,x^2y^2K^2 + y^4K^2 + 2\,x^4K + 4\,x^2y^2K + 2\,y^4K + x^4 + 2\,x^2y^2 + y^4 - 2\,x^2K^2 - 2\,y^2K^2 - 4\,x^2K - 4\,y^2K - 2\,x^2 - 2\,y^2 + K^2 + 2\,K + 1} \\
&+ \frac{4\,v^2y^4K^3}{x^4K^2 + 2\,x^2y^2K^2 + y^4K^2 + 2\,x^4K + 4\,x^2y^2K + 2\,y^4K + x^4 + 2\,x^2y^2 + y^4 - 2\,x^2K^2 - 2\,y^2K^2 - 4\,x^2K - 4\,y^2K - 2\,x^2 - 2\,y^2 + K^2 + 2\,K + 1} \\
&- \frac{lvx^4K^2}{x^4K^2 + 2\,x^2y^2K^2 + y^4K^2 + 2\,x^4K + 4\,x^2y^2K + 2\,y^4K + x^4 + 2\,x^2y^2 + y^4 - 2\,x^2K^2 - 2\,y^2K^2 - 4\,x^2K - 4\,y^2K - 2\,x^2 - 2\,y^2 + K^2 + 2\,K + 1} \\
&+ \frac{6\,v^2x^4K^2}{x^4K^2 + 2\,x^2y^2K^2 + y^4K^2 + 2\,x^4K + 4\,x^2y^2K + 2\,y^4K + x^4 + 2\,x^2y^2 + y^4 - 2\,x^2K^2 - 2\,y^2K^2 - 4\,x^2K - 4\,y^2K - 2\,x^2 - 2\,y^2 + K^2 + 2\,K + 1} \\
&- \frac{2\,lvx^2y^2K^2}{x^4K^2 + 2\,x^2y^2K^2 + y^4K^2 + 2\,x^4K + 4\,x^2y^2K + 2\,y^4K + x^4 + 2\,x^2y^2 + y^4 - 2\,x^2K^2 - 2\,y^2K^2 - 4\,x^2K - 4\,y^2K - 2\,x^2 - 2\,y^2 + K^2 + 2\,K + 1} \\
&+ \frac{12\,v^2x^2y^2K^2}{x^4K^2 + 2\,x^2y^2K^2 + y^4K^2 + 2\,x^4K + 4\,x^2y^2K + 2\,y^4K + x^4 + 2\,x^2y^2 + y^4 - 2\,x^2K^2 - 2\,y^2K^2 - 4\,x^2K - 4\,y^2K - 2\,x^2 - 2\,y^2 + K^2 + 2\,K + 1} \\
&- \frac{lvy^4K^2}{x^4K^2 + 2\,x^2y^2K^2 + y^4K^2 + 2\,x^4K + 4\,x^2y^2K + 2\,y^4K + x^4 + 2\,x^2y^2 + y^4 - 2\,x^2K^2 - 2\,y^2K^2 - 4\,x^2K - 4\,y^2K - 2\,x^2 - 2\,y^2 + K^2 + 2\,K + 1} \\
&+ \frac{6\,v^2y^4K^2}{x^4K^2 + 2\,x^2y^2K^2 + y^4K^2 + 2\,x^4K + 4\,x^2y^2K + 2\,y^4K + x^4 + 2\,x^2y^2 + y^4 - 2\,x^2K^2 - 2\,y^2K^2 - 4\,x^2K - 4\,y^2K - 2\,x^2 - 2\,y^2 + K^2 + 2\,K + 1} \\
&- \frac{2\,v^2x^2K^4}{x^4K^2 + 2\,x^2y^2K^2 + y^4K^2 + 2\,x^4K + 4\,x^2y^2K + 2\,y^4K + x^4 + 2\,x^2y^2 + y^4 - 2\,x^2K^2 - 2\,y^2K^2 - 4\,x^2K - 4\,y^2K - 2\,x^2 - 2\,y^2 + K^2 + 2\,K + 1} \\
&- \frac{2\,v^2y^2K^4}{x^4K^2 + 2\,x^2y^2K^2 + y^4K^2 + 2\,x^4K + 4\,x^2y^2K + 2\,y^4K + x^4 + 2\,x^2y^2 + y^4 - 2\,x^2K^2 - 2\,y^2K^2 - 4\,x^2K - 4\,y^2K - 2\,x^2 - 2\,y^2 + K^2 + 2\,K + 1} \\
&- \frac{2\,lvx^4K}{x^4K^2 + 2\,x^2y^2K^2 + y^4K^2 + 2\,x^4K + 4\,x^2y^2K + 2\,y^4K + x^4 + 2\,x^2y^2 + y^4 - 2\,x^2K^2 - 2\,y^2K^2 - 4\,x^2K - 4\,y^2K - 2\,x^2 - 2\,y^2 + K^2 + 2\,K + 1} \\
&+ \frac{4\,v^2x^4K}{x^4K^2 + 2\,x^2y^2K^2 + y^4K^2 + 2\,x^4K + 4\,x^2y^2K + 2\,y^4K + x^4 + 2\,x^2y^2 + y^4 - 2\,x^2K^2 - 2\,y^2K^2 - 4\,x^2K - 4\,y^2K - 2\,x^2 - 2\,y^2 + K^2 + 2\,K + 1} \\
&- \frac{4\,lvx^2y^2K}{x^4K^2 + 2\,x^2y^2K^2 + y^4K^2 + 2\,x^4K + 4\,x^2y^2K + 2\,y^4K + x^4 + 2\,x^2y^2 + y^4 - 2\,x^2K^2 - 2\,y^2K^2 - 4\,x^2K - 4\,y^2K - 2\,x^2 - 2\,y^2 + K^2 + 2\,K + 1} + \cdots
\end{aligned}
$$

$$\delta = \frac{b_x(u,x,y)}{\sqrt{|x^2+y^2-1|}|K+1|} + \frac{i\,b_y(u,x,y)}{\sqrt{|x^2+y^2-1|}|K+1|}\partial_v - \frac{x^2K^2}{2\sqrt{|x^2+y^2-1|}|K+1|} - \frac{y^2K^2}{2\sqrt{|x^2+y^2-1|}|K+1|} - \frac{x^2K}{\sqrt{|x^2+y^2-1|}|K+1|}$$
$$- \frac{y^2K}{\sqrt{|x^2+y^2-1|}|K+1|} - \frac{x^2}{2\sqrt{|x^2+y^2-1|}|K+1|} - \frac{y^2}{2\sqrt{|x^2+y^2-1|}|K+1|} + \frac{K^2}{2\sqrt{|x^2+y^2-1|}|K+1|} + \frac{K}{\sqrt{|x^2+y^2-1|}|K+1|}$$
$$+ \frac{1}{2\sqrt{|x^2+y^2-1|}|K+1|}\partial_x - \frac{i\,x^2K^2}{2\sqrt{|x^2+y^2-1|}|K+1|} - \frac{i\,y^2K^2}{2\sqrt{|x^2+y^2-1|}|K+1|} - \frac{i\,x^2K}{\sqrt{|x^2+y^2-1|}|K+1|} - \frac{i\,y^2K}{\sqrt{|x^2+y^2-1|}|K+1|}$$
$$- \frac{i\,x^2}{2\sqrt{|x^2+y^2-1|}|K+1|} - \frac{i\,y^2}{2\sqrt{|x^2+y^2-1|}|K+1|} + \frac{i\,K^2}{2\sqrt{|x^2+y^2-1|}|K+1|} + \frac{i\,K}{\sqrt{|x^2+y^2-1|}|K+1|} + \frac{i}{2\sqrt{|x^2+y^2-1|}|K+1|}\partial_y$$

$$\overline{\delta} = \frac{b_x(u,x,y)}{\sqrt{|x^2+y^2-1|}|K+1|} - \frac{i\,b_y(u,x,y)}{\sqrt{|x^2+y^2-1|}|K+1|}\partial_v - \frac{x^2K^2}{2\sqrt{|x^2+y^2-1|}|K+1|} - \frac{y^2K^2}{2\sqrt{|x^2+y^2-1|}|K+1|} - \frac{x^2K}{\sqrt{|x^2+y^2-1|}|K+1|}$$
$$- \frac{y^2K}{\sqrt{|x^2+y^2-1|}|K+1|} - \frac{x^2}{2\sqrt{|x^2+y^2-1|}|K+1|} - \frac{y^2}{2\sqrt{|x^2+y^2-1|}|K+1|} + \frac{K^2}{2\sqrt{|x^2+y^2-1|}|K+1|} + \frac{K}{\sqrt{|x^2+y^2-1|}|K+1|}$$
$$+ \frac{1}{2\sqrt{|x^2+y^2-1|}|K+1|}\partial_x + \frac{i\,x^2K^2}{2\sqrt{|x^2+y^2-1|}|K+1|} + \frac{i\,y^2K^2}{2\sqrt{|x^2+y^2-1|}|K+1|} + \frac{i\,x^2K}{\sqrt{|x^2+y^2-1|}|K+1|} + \frac{i\,y^2K}{\sqrt{|x^2+y^2-1|}|K+1|}$$
$$+ \frac{i\,x^2}{2\sqrt{|x^2+y^2-1|}|K+1|} + \frac{i\,y^2}{2\sqrt{|x^2+y^2-1|}|K+1|} - \frac{i\,K^2}{2\sqrt{|x^2+y^2-1|}|K+1|} - \frac{i\,K}{\sqrt{|x^2+y^2-1|}|K+1|} - \frac{i}{2\sqrt{|x^2+y^2-1|}|K+1|}\partial_y$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = \frac{i\,x^2K\frac{\partial}{\partial y}b_x(u,x,y)}{2(K+1)|x^2+y^2-1|} + \frac{i\,y^2K\frac{\partial}{\partial y}b_x(u,x,y)}{2(K+1)|x^2+y^2-1|} - \frac{i\,x^2K\frac{\partial}{\partial x}b_y(u,x,y)}{2(K+1)|x^2+y^2-1|} - \frac{i\,y^2K\frac{\partial}{\partial x}b_y(u,x,y)}{2(K+1)|x^2+y^2-1|} - \frac{i\,yKb_x(u,x,y)}{(K+1)|x^2+y^2-1|} + \frac{i\,xKb_y(u,x,y)}{(K+1)|x^2+y^2-1|} - \frac{x^2\frac{\partial}{\partial u}K}{(K+1)|x^2+y^2-1|}$$
$$- \frac{y^2\frac{\partial}{\partial u}K}{(K+1)|x^2+y^2-1|} + \frac{i\,x^2\frac{\partial}{\partial y}b_x(u,x,y)}{2(K+1)|x^2+y^2-1|} + \frac{i\,y^2\frac{\partial}{\partial y}b_x(u,x,y)}{2(K+1)|x^2+y^2-1|} - \frac{i\,x^2\frac{\partial}{\partial x}b_y(u,x,y)}{2(K+1)|x^2+y^2-1|} - \frac{i\,y^2\frac{\partial}{\partial x}b_y(u,x,y)}{2(K+1)|x^2+y^2-1|} - \frac{i\,yb_x(u,x,y)}{(K+1)|x^2+y^2-1|}$$
$$+ \frac{i\,xb_y(u,x,y)}{(K+1)|x^2+y^2-1|} - \frac{i\,K\frac{\partial}{\partial y}b_x(u,x,y)}{2(K+1)|x^2+y^2-1|} + \frac{i\,K\frac{\partial}{\partial x}b_y(u,x,y)}{2(K+1)|x^2+y^2-1|} + \frac{\frac{\partial}{\partial u}K}{(K+1)|x^2+y^2-1|} - \frac{i\,\frac{\partial}{\partial y}b_x(u,x,y)}{2(K+1)|x^2+y^2-1|} + \frac{i\,\frac{\partial}{\partial x}b_y(u,x,y)}{2(K+1)|x^2+y^2-1|}$$

$$\lambda = 0$$

$$\nu = \frac{x^6 K^3 \frac{\partial}{\partial x} H}{2\,\mathcal{D}_3} + \frac{3\,x^4 y^2 K^3 \frac{\partial}{\partial x} H}{2\,\mathcal{D}_3} + \frac{3\,x^2 y^4 K^3 \frac{\partial}{\partial x} H}{2\,\mathcal{D}_3} + \frac{y^6 K^3 \frac{\partial}{\partial x} H}{2\,\mathcal{D}_3} - \frac{i\,x^6 K^3 \frac{\partial}{\partial y} H}{2\,\mathcal{D}_3} - \frac{3i\,x^4 y^2 K^3 \frac{\partial}{\partial y} H}{2\,\mathcal{D}_3} - \frac{3i\,x^2 y^4 K^3 \frac{\partial}{\partial y} H}{2\,\mathcal{D}_3} - \frac{i\,y^6 K^3 \frac{\partial}{\partial y} H}{2\,\mathcal{D}_3} + \frac{2\,v x^4 K^3 b_x(u,x,y)}{\mathcal{D}_3} + \dots$$

$$\mathcal{D}_3 = \Big(x^4 K|K+1| + 2\,x^2 y^2 K|K+1| + y^4 K|K+1| + x^4|K+1| + 2\,x^2 y^2|K+1| + y^4|K+1| - 2\,x^2 K|K+1| - 2\,y^2 K|K+1| - 2\,x^2|K+1| - 2\,y^2|K+1| + K|K+1| + |K+1|\Big)\sqrt{|x^2+y^2-1|}.$$

$$\alpha = -\frac{xK^2}{4\sqrt{|x^2+y^2-1|}|K+1|} + \frac{i\,yK^2}{4\sqrt{|x^2+y^2-1|}|K+1|} - \frac{xK}{2\sqrt{|x^2+y^2-1|}|K+1|} + \frac{i\,yK}{2\sqrt{|x^2+y^2-1|}|K+1|} - \frac{x}{4\sqrt{|x^2+y^2-1|}|K+1|} + \frac{i\,y}{4\sqrt{|x^2+y^2-1|}|K+1|}$$

$$\beta = \frac{xK^2}{4\sqrt{|x^2+y^2-1|}|K+1|} + \frac{i\,yK^2}{4\sqrt{|x^2+y^2-1|}|K+1|} + \frac{xK}{2\sqrt{|x^2+y^2-1|}|K+1|} + \frac{i\,yK}{2\sqrt{|x^2+y^2-1|}|K+1|} + \frac{x}{4\sqrt{|x^2+y^2-1|}|K+1|} + \frac{i\,y}{4\sqrt{|x^2+y^2-1|}|K+1|}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = 0$$

$$\begin{aligned}\gamma = &-\frac{vK^3}{K+1} - \frac{3\,vK^2}{K+1} + \frac{i\,x^2 K \frac{\partial}{\partial y} b_x\,(u,x,y)}{4\,(K+1)|x^2+y^2-1|} + \frac{i\,y^2 K \frac{\partial}{\partial y} b_x\,(u,x,y)}{4\,(K+1)|x^2+y^2-1|} - \frac{i\,x^2 K \frac{\partial}{\partial x} b_y\,(u,x,y)}{4\,(K+1)|x^2+y^2-1|} - \frac{i\,y^2 K \frac{\partial}{\partial x} b_y\,(u,x,y)}{4\,(K+1)|x^2+y^2-1|} + \frac{lK}{2\,(K+1)} \\ &- \frac{3\,vK}{K+1} - \frac{i\,yK b_x\,(u,x,y)}{2\,(K+1)|x^2+y^2-1|} + \frac{i\,xK b_y\,(u,x,y)}{2\,(K+1)|x^2+y^2-1|} - \frac{x^2 \frac{\partial}{\partial u} K}{2\,(K+1)|x^2+y^2-1|} - \frac{y^2 \frac{\partial}{\partial u} K}{2\,(K+1)|x^2+y^2-1|} + \frac{i\,x^2 \frac{\partial}{\partial y} b_x\,(u,x,y)}{4\,(K+1)|x^2+y^2-1|} \\ &+ \frac{i\,y^2 \frac{\partial}{\partial y} b_x\,(u,x,y)}{4\,(K+1)|x^2+y^2-1|} - \frac{i\,x^2 \frac{\partial}{\partial x} b_y\,(u,x,y)}{4\,(K+1)|x^2+y^2-1|} - \frac{i\,y^2 \frac{\partial}{\partial x} b_y\,(u,x,y)}{4\,(K+1)|x^2+y^2-1|} + \frac{l}{2\,(K+1)} - \frac{v}{K+1} - \frac{i\,yb_x\,(u,x,y)}{2\,(K+1)|x^2+y^2-1|} + \frac{i\,xb_y\,(u,x,y)}{2\,(K+1)|x^2+y^2-1|} \\ &- \frac{\frac{\partial}{\partial u} K}{2\,(K+1)} - \frac{i\,K \frac{\partial}{\partial y} b_x\,(u,x,y)}{4\,(K+1)|x^2+y^2-1|} + \frac{i\,K \frac{\partial}{\partial x} b_y\,(u,x,y)}{4\,(K+1)|x^2+y^2-1|} + \frac{\frac{\partial}{\partial u} K}{2\,(K+1)|x^2+y^2-1|} - \frac{i\,\frac{\partial}{\partial y} b_x\,(u,x,y)}{4\,(K+1)|x^2+y^2-1|} + \frac{i\,\frac{\partial}{\partial x} b_y\,(u,x,y)}{4\,(K+1)|x^2+y^2-1|}\end{aligned}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\begin{aligned}\Phi_{11} = &-\frac{x^4K^2}{4\,(x^2+y^2-1)|x^2+y^2-1|} - \frac{x^2y^2K^2}{2\,(x^2+y^2-1)|x^2+y^2-1|} - \frac{y^4K^2}{4\,(x^2+y^2-1)|x^2+y^2-1|} + \frac{x^2K^2}{4\,(x^2+y^2-1)} + \frac{y^2K^2}{4\,(x^2+y^2-1)} \\ &- \frac{x^4K}{2\,(x^2+y^2-1)|x^2+y^2-1|} - \frac{x^2y^2K}{(x^2+y^2-1)|x^2+y^2-1|} - \frac{y^4K}{2\,(x^2+y^2-1)|x^2+y^2-1|} + \frac{x^2K}{2\,(x^2+y^2-1)} + \frac{y^2K}{2\,(x^2+y^2-1)} - \frac{x^4}{4\,(x^2+y^2-1)|x^2+y^2-1|} \\ &- \frac{x^2y^2}{2\,(x^2+y^2-1)|x^2+y^2-1|} - \frac{y^4}{4\,(x^2+y^2-1)|x^2+y^2-1|} + \frac{3\,x^2K^2}{8\,(x^2+y^2-1)|x^2+y^2-1|} + \frac{3\,y^2K^2}{8\,(x^2+y^2-1)|x^2+y^2-1|} + \frac{x^2}{4\,(x^2+y^2-1)} \\ &+ \frac{y^2}{4\,(x^2+y^2-1)} - \frac{K^2}{8\,(x^2+y^2-1)} + \frac{3\,x^2K}{4\,(x^2+y^2-1)|x^2+y^2-1|} + \frac{3\,y^2K}{4\,(x^2+y^2-1)|x^2+y^2-1|} - \frac{K}{4\,(x^2+y^2-1)} + \frac{3\,x^2}{8\,(x^2+y^2-1)|x^2+y^2-1|} \\ &+ \frac{3\,y^2}{8\,(x^2+y^2-1)|x^2+y^2-1|} - \frac{K^2}{8\,(x^2+y^2-1)|x^2+y^2-1|} - \frac{1}{8\,(x^2+y^2-1)} - \frac{K}{4\,(x^2+y^2-1)|x^2+y^2-1|} - \frac{1}{8\,(x^2+y^2-1)|x^2+y^2-1|}\end{aligned}$$

$$
\begin{aligned}
\Phi_{12} = {} & \frac{i\,x^4K^2\frac{\partial^2}{\partial x\partial y}b_x\left(u,x,y\right)}{8\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} + \frac{i\,x^2y^2K^2\frac{\partial^2}{\partial x\partial y}b_x\left(u,x,y\right)}{4\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} + \frac{i\,y^4K^2\frac{\partial^2}{\partial x\partial y}b_x\left(u,x,y\right)}{8\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} \\
& - \frac{x^4K^2\frac{\partial^2}{(\partial y)^2}b_x\left(u,x,y\right)}{8\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} - \frac{x^2y^2K^2\frac{\partial^2}{(\partial y)^2}b_x\left(u,x,y\right)}{4\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} - \frac{y^4K^2\frac{\partial^2}{(\partial y)^2}b_x\left(u,x,y\right)}{8\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} \\
& - \frac{i\,x^4K^2\frac{\partial^2}{(\partial x)^2}b_y\left(u,x,y\right)}{8\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} - \frac{i\,x^2y^2K^2\frac{\partial^2}{(\partial x)^2}b_y\left(u,x,y\right)}{4\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} - \frac{i\,y^4K^2\frac{\partial^2}{(\partial x)^2}b_y\left(u,x,y\right)}{8\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} \\
& + \frac{x^4K^2\frac{\partial^2}{\partial x\partial y}b_y\left(u,x,y\right)}{8\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} + \frac{x^2y^2K^2\frac{\partial^2}{\partial x\partial y}b_y\left(u,x,y\right)}{4\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} + \frac{y^4K^2\frac{\partial^2}{\partial x\partial y}b_y\left(u,x,y\right)}{8\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} \\
& - \frac{i\,x^2yK^2\frac{\partial}{\partial x}b_x\left(u,x,y\right)}{4\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} - \frac{i\,y^3K^2\frac{\partial}{\partial x}b_x\left(u,x,y\right)}{4\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} + \frac{i\,x^4K\frac{\partial^2}{\partial x\partial y}b_x\left(u,x,y\right)}{4\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} \\
& + \frac{i\,x^2y^2K\frac{\partial^2}{\partial x\partial y}b_x\left(u,x,y\right)}{2\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} + \frac{i\,y^4K\frac{\partial^2}{\partial x\partial y}b_x\left(u,x,y\right)}{4\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} + \frac{x^2yK^2\frac{\partial}{\partial y}b_x\left(u,x,y\right)}{4\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} \\
& + \frac{y^3K^2\frac{\partial}{\partial y}b_x\left(u,x,y\right)}{4\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} - \frac{x^4K\frac{\partial^2}{(\partial y)^2}b_x\left(u,x,y\right)}{4\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} - \frac{x^2y^2K\frac{\partial^2}{(\partial y)^2}b_x\left(u,x,y\right)}{2\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} - \cdots
\end{aligned}
$$

$$\Phi_{20} = 0$$

$$
\begin{aligned}
\Phi_{21} = {} & -\frac{i\,x^4K^2\frac{\partial^2}{\partial x\partial y}b_x\left(u,x,y\right)}{8\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} - \frac{i\,x^2y^2K^2\frac{\partial^2}{\partial x\partial y}b_x\left(u,x,y\right)}{4\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} - \frac{i\,y^4K^2\frac{\partial^2}{\partial x\partial y}b_x\left(u,x,y\right)}{8\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} \\
& - \frac{x^4K^2\frac{\partial^2}{(\partial y)^2}b_x\left(u,x,y\right)}{8\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} - \frac{x^2y^2K^2\frac{\partial^2}{(\partial y)^2}b_x\left(u,x,y\right)}{4\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} - \frac{y^4K^2\frac{\partial^2}{(\partial y)^2}b_x\left(u,x,y\right)}{8\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} \\
& + \frac{i\,x^4K^2\frac{\partial^2}{(\partial x)^2}b_y\left(u,x,y\right)}{8\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} + \frac{i\,x^2y^2K^2\frac{\partial^2}{(\partial x)^2}b_y\left(u,x,y\right)}{4\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} + \frac{i\,y^4K^2\frac{\partial^2}{(\partial x)^2}b_y\left(u,x,y\right)}{8\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} \\
& + \frac{x^4K^2\frac{\partial^2}{\partial x\partial y}b_y\left(u,x,y\right)}{8\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} + \frac{x^2y^2K^2\frac{\partial^2}{\partial x\partial y}b_y\left(u,x,y\right)}{4\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} + \frac{y^4K^2\frac{\partial^2}{\partial x\partial y}b_y\left(u,x,y\right)}{8\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} \\
& + \frac{i\,x^2yK^2\frac{\partial}{\partial x}b_x\left(u,x,y\right)}{4\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} + \frac{i\,y^3K^2\frac{\partial}{\partial x}b_x\left(u,x,y\right)}{4\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} - \frac{i\,x^4K\frac{\partial^2}{\partial x\partial y}b_x\left(u,x,y\right)}{4\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} \\
& - \frac{i\,x^2y^2K\frac{\partial^2}{\partial x\partial y}b_x\left(u,x,y\right)}{2\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} - \frac{i\,y^4K\frac{\partial^2}{\partial x\partial y}b_x\left(u,x,y\right)}{4\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} + \frac{x^2yK^2\frac{\partial}{\partial y}b_x\left(u,x,y\right)}{4\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} \\
& + \frac{y^3K^2\frac{\partial}{\partial y}b_x\left(u,x,y\right)}{4\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} - \frac{x^4K\frac{\partial^2}{(\partial y)^2}b_x\left(u,x,y\right)}{4\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} - \frac{x^2y^2K\frac{\partial^2}{(\partial y)^2}b_x\left(u,x,y\right)}{2\left(x^2|K+1|+y^2|K+1|-|K+1|\right)\sqrt{|x^2+y^2-1|}} - \cdots
\end{aligned}
$$

$$\Phi_{22} = \frac{x^8 K^4 \frac{\partial^2}{(\partial x)^2} H}{4\,\mathcal{D}_4} + \frac{x^6 y^2 K^4 \frac{\partial^2}{(\partial x)^2} H}{\mathcal{D}_4} + \frac{3\,x^4 y^4 K^4 \frac{\partial^2}{(\partial x)^2} H}{2\,\mathcal{D}_4} + \frac{x^2 y^6 K^4 \frac{\partial^2}{(\partial x)^2} H}{\mathcal{D}_4} + \frac{y^8 K^4 \frac{\partial^2}{(\partial x)^2} H}{4\,\mathcal{D}_4} + \frac{x^8 K^4 \frac{\partial^2}{(\partial y)^2} H}{4\,\mathcal{D}_4} + \frac{x^6 y^2 K^4 \frac{\partial^2}{(\partial y)^2} H}{\mathcal{D}_4} + \frac{3\,x^4 y^4 K^4 \frac{\partial^2}{(\partial y)^2} H}{2\,\mathcal{D}_4} + \frac{x^2 y^6 K^4 \frac{\partial^2}{(\partial y)^2} H}{\mathcal{D}_4} + \dots$$

$$\begin{aligned}\mathcal{D}_4 = {} & x^6 K^2 + 3\,x^4 y^2 K^2 + 3\,x^2 y^4 K^2 + y^6 K^2 + 2\,x^6 K + 6\,x^4 y^2 K + 6\,x^2 y^4 K + 2\,y^6 K + x^6 + 3\,x^4 y^2 + 3\,x^2 y^4 + y^6 - 3\,x^4 K^2 - 6\,x^2 y^2 K^2 - 3\,y^4 K^2 - 6\,x^4 K - 12\,x^2 y^2 K - 6\,y^4 K \\ & - 3\,x^4 - 6\,x^2 y^2 - 3\,y^4 + 3\,x^2 K^2 + 3\,y^2 K^2 + 6\,x^2 K + 6\,y^2 K + 3\,x^2 + 3\,y^2 - K^2 - 2\,K - 1.\end{aligned}$$

$$\Lambda = -\frac{x^2 K^2}{6\,(x^2+y^2-1)} - \frac{y^2 K^2}{6\,(x^2+y^2-1)} - \frac{x^2 K}{3\,(x^2+y^2-1)} - \frac{y^2 K}{3\,(x^2+y^2-1)} - \frac{x^2}{6\,(x^2+y^2-1)} - \frac{y^2}{6\,(x^2+y^2-1)} + \frac{K^2}{4\,(x^2+y^2-1)} + \frac{K}{2\,(x^2+y^2-1)} + \frac{1}{4\,(x^2+y^2-1)}$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = -\frac{x^2 K^2}{3\,|x^2+y^2-1|} - \frac{y^2 K^2}{3\,|x^2+y^2-1|} - \frac{2\,x^2 K}{3\,|x^2+y^2-1|} - \frac{2\,y^2 K}{3\,|x^2+y^2-1|} - \frac{x^2}{3\,|x^2+y^2-1|} - \frac{y^2}{3\,|x^2+y^2-1|} + \frac{K^2}{2\,|x^2+y^2-1|} + \frac{K}{|x^2+y^2-1|} + \frac{1}{2\,|x^2+y^2-1|}$$

$$\begin{aligned}\Psi_3 = {} & \frac{i\,x^4 K^2 \frac{\partial^2}{\partial x \partial y} b_x(u,x,y)}{8\,(x^2|K+1| + y^2|K+1| - |K+1|)\sqrt{|x^2+y^2-1|}} + \frac{i\,x^2 y^2 K^2 \frac{\partial^2}{\partial x \partial y} b_x(u,x,y)}{4\,(x^2|K+1| + y^2|K+1| - |K+1|)\sqrt{|x^2+y^2-1|}} + \frac{i\,y^4 K^2 \frac{\partial^2}{\partial x \partial y} b_x(u,x,y)}{8\,(x^2|K+1| + y^2|K+1| - |K+1|)\sqrt{|x^2+y^2-1|}} \\ & + \frac{x^4 K^2 \frac{\partial^2}{(\partial y)^2} b_x(u,x,y)}{8\,(x^2|K+1| + y^2|K+1| - |K+1|)\sqrt{|x^2+y^2-1|}} + \frac{x^2 y^2 K^2 \frac{\partial^2}{(\partial y)^2} b_x(u,x,y)}{4\,(x^2|K+1| + y^2|K+1| - |K+1|)\sqrt{|x^2+y^2-1|}} + \frac{y^4 K^2 \frac{\partial^2}{(\partial y)^2} b_x(u,x,y)}{8\,(x^2|K+1| + y^2|K+1| - |K+1|)\sqrt{|x^2+y^2-1|}} \\ & - \frac{i\,x^4 K^2 \frac{\partial^2}{(\partial x)^2} b_y(u,x,y)}{8\,(x^2|K+1| + y^2|K+1| - |K+1|)\sqrt{|x^2+y^2-1|}} - \frac{i\,x^2 y^2 K^2 \frac{\partial^2}{(\partial x)^2} b_y(u,x,y)}{4\,(x^2|K+1| + y^2|K+1| - |K+1|)\sqrt{|x^2+y^2-1|}} - \frac{i\,y^4 K^2 \frac{\partial^2}{(\partial x)^2} b_y(u,x,y)}{8\,(x^2|K+1| + y^2|K+1| - |K+1|)\sqrt{|x^2+y^2-1|}} \\ & - \frac{x^4 K^2 \frac{\partial^2}{\partial x \partial y} b_y(u,x,y)}{8\,(x^2|K+1| + y^2|K+1| - |K+1|)\sqrt{|x^2+y^2-1|}} - \frac{x^2 y^2 K^2 \frac{\partial^2}{\partial x \partial y} b_y(u,x,y)}{4\,(x^2|K+1| + y^2|K+1| - |K+1|)\sqrt{|x^2+y^2-1|}} - \frac{y^4 K^2 \frac{\partial^2}{\partial x \partial y} b_y(u,x,y)}{8\,(x^2|K+1| + y^2|K+1| - |K+1|)\sqrt{|x^2+y^2-1|}} \\ & - \frac{i\,x^2 y K^2 \frac{\partial}{\partial x} b_x(u,x,y)}{4\,(x^2|K+1| + y^2|K+1| - |K+1|)\sqrt{|x^2+y^2-1|}} - \frac{i\,y^3 K^2 \frac{\partial}{\partial x} b_x(u,x,y)}{4\,(x^2|K+1| + y^2|K+1| - |K+1|)\sqrt{|x^2+y^2-1|}} + \frac{i\,x^4 K \frac{\partial^2}{\partial x \partial y} b_x(u,x,y)}{4\,(x^2|K+1| + y^2|K+1| - |K+1|)\sqrt{|x^2+y^2-1|}} \\ & + \frac{i\,x^2 y^2 K \frac{\partial^2}{\partial x \partial y} b_x(u,x,y)}{2\,(x^2|K+1| + y^2|K+1| - |K+1|)\sqrt{|x^2+y^2-1|}} + \frac{i\,y^4 K \frac{\partial^2}{\partial x \partial y} b_x(u,x,y)}{4\,(x^2|K+1| + y^2|K+1| - |K+1|)\sqrt{|x^2+y^2-1|}} - \frac{x^2 y K^2 \frac{\partial}{\partial y} b_x(u,x,y)}{4\,(x^2|K+1| + y^2|K+1| - |K+1|)\sqrt{|x^2+y^2-1|}} \\ & - \frac{y^3 K^2 \frac{\partial}{\partial y} b_x(u,x,y)}{4\,(x^2|K+1| + y^2|K+1| - |K+1|)\sqrt{|x^2+y^2-1|}} + \frac{x^4 K \frac{\partial^2}{(\partial y)^2} b_x(u,x,y)}{4\,(x^2|K+1| + y^2|K+1| - |K+1|)\sqrt{|x^2+y^2-1|}} + \frac{x^2 y^2 K \frac{\partial^2}{(\partial y)^2} b_x(u,x,y)}{2\,(x^2|K+1| + y^2|K+1| - |K+1|)\sqrt{|x^2+y^2-1|}} + \dots\end{aligned}$$

$$\Psi_4 = -\frac{x^8 K^3 \frac{\partial^2}{(\partial x)^2} H}{4\,\mathcal{D}_5} - \frac{x^6 y^2 K^3 \frac{\partial^2}{(\partial x)^2} H}{\mathcal{D}_5} - \frac{3\,x^4 y^4 K^3 \frac{\partial^2}{(\partial x)^2} H}{2\,\mathcal{D}_5} - \frac{x^2 y^6 K^3 \frac{\partial^2}{(\partial x)^2} H}{\mathcal{D}_5} - \frac{y^8 K^3 \frac{\partial^2}{(\partial x)^2} H}{4\,\mathcal{D}_5} + \frac{i\,x^8 K^3 \frac{\partial^2}{\partial x \partial y} H}{2\,\mathcal{D}_5} + \frac{2i\,x^6 y^2 K^3 \frac{\partial^2}{\partial x \partial y} H}{\mathcal{D}_5} + \frac{3i\,x^4 y^4 K^3 \frac{\partial^2}{\partial x \partial y} H}{\mathcal{D}_5} + \frac{2i\,x^2 y^6 K^3 \frac{\partial^2}{\partial x \partial y} H}{\mathcal{D}_5}$$
$$+ \frac{i\,y^8 K^3 \frac{\partial^2}{\partial x \partial y} H}{2\,\mathcal{D}_5} + \frac{x^8 K^3 \frac{\partial^2}{(\partial y)^2} H}{4\,\mathcal{D}_5} + \frac{x^6 y^2 K^3 \frac{\partial^2}{(\partial y)^2} H}{\mathcal{D}_5} + \frac{3\,x^4 y^4 K^3 \frac{\partial^2}{(\partial y)^2} H}{2\,\mathcal{D}_5} + \frac{x^2 y^6 K^3 \frac{\partial^2}{(\partial y)^2} H}{\mathcal{D}_5} + \frac{y^8 K^3 \frac{\partial^2}{(\partial y)^2} H}{4\,\mathcal{D}_5} - \frac{x^7 K^3 \frac{\partial}{\partial x} H}{2\,\mathcal{D}_5} + \frac{i\,x^6 y K^3 \frac{\partial}{\partial x} H}{2\,\mathcal{D}_5} - \frac{3\,x^5 y^2 K^3 \frac{\partial}{\partial x} H}{2\,\mathcal{D}_5} + \frac{3i\,x^4 y^3 K^3 \frac{\partial}{\partial x} H}{2\,\mathcal{D}_5} - \dots$$

$$\mathcal{D}_5 = \Big(x^4 K + 2\,x^2 y^2 K + y^4 K + x^4 + 2\,x^2 y^2 + y^4 - 2\,x^2 K - 2\,y^2 K - 2\,x^2 - 2\,y^2 + K + 1\Big)\big|x^2 + y^2 - 1\big|.$$

Newman-Penrose Equations

$$NP6 = -\frac{7\,x^4 K^2}{12\,(x^2+y^2-1)|x^2+y^2-1|} - \frac{7\,x^2 y^2 K^2}{6\,(x^2+y^2-1)|x^2+y^2-1|} - \frac{7\,y^4 K^2}{12\,(x^2+y^2-1)|x^2+y^2-1|} - \frac{7\,x^2 K^2}{12\,(x^2+y^2-1)} - \frac{7\,y^2 K^2}{12\,(x^2+y^2-1)}$$
$$- \frac{7\,x^4 K}{6\,(x^2+y^2-1)|x^2+y^2-1|} - \frac{7\,x^2 y^2 K}{3\,(x^2+y^2-1)|x^2+y^2-1|} - \frac{7\,y^4 K}{6\,(x^2+y^2-1)|x^2+y^2-1|} - \frac{7\,x^2 K}{6\,(x^2+y^2-1)} - \frac{7\,y^2 K}{6\,(x^2+y^2-1)} - \frac{7\,x^4}{12\,(x^2+y^2-1)|x^2+y^2-1|}$$
$$- \frac{7\,x^2 y^2}{6\,(x^2+y^2-1)|x^2+y^2-1|} - \frac{7\,y^4}{12\,(x^2+y^2-1)|x^2+y^2-1|} + \frac{29\,x^2 K^2}{24\,(x^2+y^2-1)|x^2+y^2-1|} + \frac{29\,y^2 K^2}{24\,(x^2+y^2-1)|x^2+y^2-1|} - \frac{7\,x^2}{12\,(x^2+y^2-1)}$$
$$- \frac{7\,y^2}{12\,(x^2+y^2-1)} + \frac{5\,K^2}{8\,(x^2+y^2-1)} + \frac{29\,x^2 K}{12\,(x^2+y^2-1)|x^2+y^2-1|} + \frac{29\,y^2 K}{12\,(x^2+y^2-1)|x^2+y^2-1|} + \frac{5\,K}{4\,(x^2+y^2-1)} + \frac{29\,x^2}{24\,(x^2+y^2-1)|x^2+y^2-1|}$$
$$+ \frac{29\,y^2}{24\,(x^2+y^2-1)|x^2+y^2-1|} - \frac{5\,K^2}{8\,(x^2+y^2-1)|x^2+y^2-1|} + \frac{5}{8\,(x^2+y^2-1)} - \frac{5\,K}{4\,(x^2+y^2-1)|x^2+y^2-1|} - \frac{5}{8\,(x^2+y^2-1)|x^2+y^2-1|}$$

$$NP8 = -\frac{x^4 K^2}{3\,(x^2+y^2-1)|x^2+y^2-1|} - \frac{2\,x^2 y^2 K^2}{3\,(x^2+y^2-1)|x^2+y^2-1|} - \frac{y^4 K^2}{3\,(x^2+y^2-1)|x^2+y^2-1|} - \frac{x^2 K^2}{3\,(x^2+y^2-1)} - \frac{y^2 K^2}{3\,(x^2+y^2-1)} - \frac{2\,x^4 K}{3\,(x^2+y^2-1)|x^2+y^2-1|}$$
$$- \frac{4\,x^2 y^2 K}{3\,(x^2+y^2-1)|x^2+y^2-1|} - \frac{2\,y^4 K}{3\,(x^2+y^2-1)|x^2+y^2-1|} - \frac{2\,x^2 K}{3\,(x^2+y^2-1)} - \frac{2\,y^2 K}{3\,(x^2+y^2-1)} - \frac{x^4}{3\,(x^2+y^2-1)|x^2+y^2-1|}$$
$$- \frac{2\,x^2 y^2}{3\,(x^2+y^2-1)|x^2+y^2-1|} - \frac{y^4}{3\,(x^2+y^2-1)|x^2+y^2-1|} + \frac{5\,x^2 K^2}{6\,(x^2+y^2-1)|x^2+y^2-1|} + \frac{5\,y^2 K^2}{6\,(x^2+y^2-1)|x^2+y^2-1|} - \frac{x^2}{3\,(x^2+y^2-1)}$$
$$- \frac{y^2}{3\,(x^2+y^2-1)} + \frac{K^2}{2\,(x^2+y^2-1)} + \frac{5\,x^2 K}{3\,(x^2+y^2-1)|x^2+y^2-1|} + \frac{5\,y^2 K}{3\,(x^2+y^2-1)|x^2+y^2-1|} + \frac{K}{x^2+y^2-1} + \frac{5\,x^2}{6\,(x^2+y^2-1)|x^2+y^2-1|}$$
$$+ \frac{5\,y^2}{6\,(x^2+y^2-1)|x^2+y^2-1|} - \frac{K^2}{2\,(x^2+y^2-1)|x^2+y^2-1|} + \frac{1}{2\,(x^2+y^2-1)} - \frac{K}{(x^2+y^2-1)|x^2+y^2-1|} - \frac{1}{2\,(x^2+y^2-1)|x^2+y^2-1|}$$

$$
\begin{aligned}
NP12 = {} & \frac{x^4K^2}{12\left(x^2+y^2-1\right)\left|x^2+y^2-1\right|} + \frac{x^2y^2K^2}{6\left(x^2+y^2-1\right)\left|x^2+y^2-1\right|} + \frac{y^4K^2}{12\left(x^2+y^2-1\right)\left|x^2+y^2-1\right|} + \frac{x^2K^2}{12\left(x^2+y^2-1\right)} + \frac{y^2K^2}{12\left(x^2+y^2-1\right)} \\
& + \frac{x^4K}{6\left(x^2+y^2-1\right)\left|x^2+y^2-1\right|} + \frac{x^2y^2K}{3\left(x^2+y^2-1\right)\left|x^2+y^2-1\right|} + \frac{y^4K}{6\left(x^2+y^2-1\right)\left|x^2+y^2-1\right|} + \frac{x^2K}{6\left(x^2+y^2-1\right)} + \frac{y^2K}{6\left(x^2+y^2-1\right)} + \frac{x^4}{12\left(x^2+y^2-1\right)\left|x^2+y^2-1\right|} \\
& + \frac{x^2y^2}{6\left(x^2+y^2-1\right)\left|x^2+y^2-1\right|} + \frac{y^4}{12\left(x^2+y^2-1\right)\left|x^2+y^2-1\right|} + \frac{x^2K^2}{24\left(x^2+y^2-1\right)\left|x^2+y^2-1\right|} + \frac{y^2K^2}{24\left(x^2+y^2-1\right)\left|x^2+y^2-1\right|} + \frac{x^2}{12\left(x^2+y^2-1\right)} \\
& + \frac{y^2}{12\left(x^2+y^2-1\right)} + \frac{K^2}{8\left(x^2+y^2-1\right)} + \frac{x^2K}{12\left(x^2+y^2-1\right)\left|x^2+y^2-1\right|} + \frac{y^2K}{12\left(x^2+y^2-1\right)\left|x^2+y^2-1\right|} + \frac{K}{4\left(x^2+y^2-1\right)} + \frac{x^2}{24\left(x^2+y^2-1\right)\left|x^2+y^2-1\right|} \\
& + \frac{y^2}{24\left(x^2+y^2-1\right)\left|x^2+y^2-1\right|} - \frac{K^2}{8\left(x^2+y^2-1\right)\left|x^2+y^2-1\right|} + \frac{1}{8\left(x^2+y^2-1\right)} - \frac{K}{4\left(x^2+y^2-1\right)\left|x^2+y^2-1\right|} - \frac{1}{8\left(x^2+y^2-1\right)\left|x^2+y^2-1\right|}
\end{aligned}
$$

$$
\begin{aligned}
NP13 = &-\frac{i\,x^6K^2\frac{\partial^2}{\partial x\partial y}b_x\left(u,x,y\right)}{4\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}}\\
&-\frac{3i\,x^4y^2K^2\frac{\partial^2}{\partial x\partial y}b_x\left(u,x,y\right)}{4\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}}\\
&-\frac{3i\,x^2y^4K^2\frac{\partial^2}{\partial x\partial y}b_x\left(u,x,y\right)}{4\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}}\\
&-\frac{i\,y^6K^2\frac{\partial^2}{\partial x\partial y}b_x\left(u,x,y\right)}{4\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}}\\
&-\frac{x^6K^2\frac{\partial^2}{(\partial y)^2}b_x\left(u,x,y\right)}{4\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}}\\
&-\frac{3\,x^4y^2K^2\frac{\partial^2}{(\partial y)^2}b_x\left(u,x,y\right)}{4\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}}\\
&-\frac{3\,x^2y^4K^2\frac{\partial^2}{(\partial y)^2}b_x\left(u,x,y\right)}{4\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}}\\
&-\frac{y^6K^2\frac{\partial^2}{(\partial y)^2}b_x\left(u,x,y\right)}{4\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}}\\
&+\frac{i\,x^6K^2\frac{\partial^2}{(\partial x)^2}b_y\left(u,x,y\right)}{4\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}}\\
&+\frac{3i\,x^4y^2K^2\frac{\partial^2}{(\partial x)^2}b_y\left(u,x,y\right)}{4\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}}\\
&+\frac{3i\,x^2y^4K^2\frac{\partial^2}{(\partial x)^2}b_y\left(u,x,y\right)}{4\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}}\\
&+\frac{i\,y^6K^2\frac{\partial^2}{(\partial x)^2}b_y\left(u,x,y\right)}{4\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}}\\
&+\frac{x^6K^2\frac{\partial^2}{\partial x\partial y}b_y\left(u,x,y\right)}{4\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}}\\
&+\frac{3\,x^4y^2K^2\frac{\partial^2}{\partial x\partial y}b_y\left(u,x,y\right)}{4\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}}+\ldots
\end{aligned}
$$

$NP14$

$$= \frac{x^{10}K^3\frac{\partial^2}{(\partial x)^2}H}{4\,(x^6K+3\,x^4y^2K+3\,x^2y^4K+y^6K+x^6+3\,x^4y^2+3\,x^2y^4+y^6-3\,x^4K-6\,x^2y^2K-3\,y^4K-3\,x^4-6\,x^2y^2-3\,y^4+3\,x^2K+3\,y^2K+3\,x^2+3\,y^2-K-1)|x^2+y^2-1|}$$

$$+ \frac{5\,x^8y^2K^3\frac{\partial^2}{(\partial x)^2}H}{4\,(x^6K+3\,x^4y^2K+3\,x^2y^4K+y^6K+x^6+3\,x^4y^2+3\,x^2y^4+y^6-3\,x^4K-6\,x^2y^2K-3\,y^4K-3\,x^4-6\,x^2y^2-3\,y^4+3\,x^2K+3\,y^2K+3\,x^2+3\,y^2-K-1)|x^2+y^2-1|}$$

$$+ \frac{5\,x^6y^4K^3\frac{\partial^2}{(\partial x)^2}H}{2\,(x^6K+3\,x^4y^2K+3\,x^2y^4K+y^6K+x^6+3\,x^4y^2+3\,x^2y^4+y^6-3\,x^4K-6\,x^2y^2K-3\,y^4K-3\,x^4-6\,x^2y^2-3\,y^4+3\,x^2K+3\,y^2K+3\,x^2+3\,y^2-K-1)|x^2+y^2-1|}$$

$$+ \frac{5\,x^4y^6K^3\frac{\partial^2}{(\partial x)^2}H}{2\,(x^6K+3\,x^4y^2K+3\,x^2y^4K+y^6K+x^6+3\,x^4y^2+3\,x^2y^4+y^6-3\,x^4K-6\,x^2y^2K-3\,y^4K-3\,x^4-6\,x^2y^2-3\,y^4+3\,x^2K+3\,y^2K+3\,x^2+3\,y^2-K-1)|x^2+y^2-1|}$$

$$+ \frac{5\,x^2y^8K^3\frac{\partial^2}{(\partial x)^2}H}{4\,(x^6K+3\,x^4y^2K+3\,x^2y^4K+y^6K+x^6+3\,x^4y^2+3\,x^2y^4+y^6-3\,x^4K-6\,x^2y^2K-3\,y^4K-3\,x^4-6\,x^2y^2-3\,y^4+3\,x^2K+3\,y^2K+3\,x^2+3\,y^2-K-1)|x^2+y^2-1|}$$

$$+ \frac{y^{10}K^3\frac{\partial^2}{(\partial x)^2}H}{4\,(x^6K+3\,x^4y^2K+3\,x^2y^4K+y^6K+x^6+3\,x^4y^2+3\,x^2y^4+y^6-3\,x^4K-6\,x^2y^2K-3\,y^4K-3\,x^4-6\,x^2y^2-3\,y^4+3\,x^2K+3\,y^2K+3\,x^2+3\,y^2-K-1)|x^2+y^2-1|}$$

$$+ \frac{x^{10}K^3\frac{\partial^2}{(\partial y)^2}H}{4\,(x^6K+3\,x^4y^2K+3\,x^2y^4K+y^6K+x^6+3\,x^4y^2+3\,x^2y^4+y^6-3\,x^4K-6\,x^2y^2K-3\,y^4K-3\,x^4-6\,x^2y^2-3\,y^4+3\,x^2K+3\,y^2K+3\,x^2+3\,y^2-K-1)|x^2+y^2-1|}$$

$$+ \frac{5\,x^8y^2K^3\frac{\partial^2}{(\partial y)^2}H}{4\,(x^6K+3\,x^4y^2K+3\,x^2y^4K+y^6K+x^6+3\,x^4y^2+3\,x^2y^4+y^6-3\,x^4K-6\,x^2y^2K-3\,y^4K-3\,x^4-6\,x^2y^2-3\,y^4+3\,x^2K+3\,y^2K+3\,x^2+3\,y^2-K-1)|x^2+y^2-1|}$$

$$+ \frac{5\,x^6y^4K^3\frac{\partial^2}{(\partial y)^2}H}{2\,(x^6K+3\,x^4y^2K+3\,x^2y^4K+y^6K+x^6+3\,x^4y^2+3\,x^2y^4+y^6-3\,x^4K-6\,x^2y^2K-3\,y^4K-3\,x^4-6\,x^2y^2-3\,y^4+3\,x^2K+3\,y^2K+3\,x^2+3\,y^2-K-1)|x^2+y^2-1|}$$

$$+ \frac{5\,x^4y^6K^3\frac{\partial^2}{(\partial y)^2}H}{2\,(x^6K+3\,x^4y^2K+3\,x^2y^4K+y^6K+x^6+3\,x^4y^2+3\,x^2y^4+y^6-3\,x^4K-6\,x^2y^2K-3\,y^4K-3\,x^4-6\,x^2y^2-3\,y^4+3\,x^2K+3\,y^2K+3\,x^2+3\,y^2-K-1)|x^2+y^2-1|}$$

$$+ \frac{5\,x^2y^8K^3\frac{\partial^2}{(\partial y)^2}H}{4\,(x^6K+3\,x^4y^2K+3\,x^2y^4K+y^6K+x^6+3\,x^4y^2+3\,x^2y^4+y^6-3\,x^4K-6\,x^2y^2K-3\,y^4K-3\,x^4-6\,x^2y^2-3\,y^4+3\,x^2K+3\,y^2K+3\,x^2+3\,y^2-K-1)|x^2+y^2-1|}$$

$$+ \frac{y^{10}K^3\frac{\partial^2}{(\partial y)^2}H}{4\,(x^6K+3\,x^4y^2K+3\,x^2y^4K+y^6K+x^6+3\,x^4y^2+3\,x^2y^4+y^6-3\,x^4K-6\,x^2y^2K-3\,y^4K-3\,x^4-6\,x^2y^2-3\,y^4+3\,x^2K+3\,y^2K+3\,x^2+3\,y^2-K-1)|x^2+y^2-1|}$$

$$+ \ldots$$

$$
\begin{aligned}
NP15 = & \frac{i\,x^6K^2\frac{\partial^2}{\partial x\partial y}b_x\left(u,x,y\right)}{8\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}} \\
& + \frac{3i\,x^4y^2K^2\frac{\partial^2}{\partial x\partial y}b_x\left(u,x,y\right)}{8\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}} \\
& + \frac{3i\,x^2y^4K^2\frac{\partial^2}{\partial x\partial y}b_x\left(u,x,y\right)}{8\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}} \\
& + \frac{i\,y^6K^2\frac{\partial^2}{\partial x\partial y}b_x\left(u,x,y\right)}{8\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}} \\
& - \frac{x^6K^2\frac{\partial^2}{(\partial y)^2}b_x\left(u,x,y\right)}{8\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}} \\
& - \frac{3\,x^4y^2K^2\frac{\partial^2}{(\partial y)^2}b_x\left(u,x,y\right)}{8\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}} \\
& - \frac{3\,x^2y^4K^2\frac{\partial^2}{(\partial y)^2}b_x\left(u,x,y\right)}{8\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}} \\
& - \frac{y^6K^2\frac{\partial^2}{(\partial y)^2}b_x\left(u,x,y\right)}{8\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}} \\
& - \frac{i\,x^6K^2\frac{\partial^2}{(\partial x)^2}b_y\left(u,x,y\right)}{8\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}} \\
& - \frac{3i\,x^4y^2K^2\frac{\partial^2}{(\partial x)^2}b_y\left(u,x,y\right)}{8\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}} \\
& - \frac{3i\,x^2y^4K^2\frac{\partial^2}{(\partial x)^2}b_y\left(u,x,y\right)}{8\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}} \\
& - \frac{i\,y^6K^2\frac{\partial^2}{(\partial x)^2}b_y\left(u,x,y\right)}{8\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}} \\
& + \frac{x^6K^2\frac{\partial^2}{\partial x\partial y}b_y\left(u,x,y\right)}{8\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}} \\
& + \frac{3\,x^4y^2K^2\frac{\partial^2}{\partial x\partial y}b_y\left(u,x,y\right)}{8\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}} + \ldots
\end{aligned}
$$

$$
\begin{aligned}
NP17 = {} & \frac{x^4K^2}{3\,(x^2+y^2-1)|x^2+y^2-1|} + \frac{2\,x^2y^2K^2}{3\,(x^2+y^2-1)|x^2+y^2-1|} + \frac{y^4K^2}{3\,(x^2+y^2-1)|x^2+y^2-1|} + \frac{x^2K^2}{3\,(x^2+y^2-1)} + \frac{y^2K^2}{3\,(x^2+y^2-1)} + \frac{2\,x^4K}{3\,(x^2+y^2-1)|x^2+y^2-1|} \\
& + \frac{4\,x^2y^2K}{3\,(x^2+y^2-1)|x^2+y^2-1|} + \frac{2\,y^4K}{3\,(x^2+y^2-1)|x^2+y^2-1|} + \frac{2\,x^2K}{3\,(x^2+y^2-1)} + \frac{2\,y^2K}{3\,(x^2+y^2-1)} + \frac{x^4}{3\,(x^2+y^2-1)|x^2+y^2-1|} \\
& + \frac{2\,x^2y^2}{3\,(x^2+y^2-1)|x^2+y^2-1|} + \frac{y^4}{3\,(x^2+y^2-1)|x^2+y^2-1|} - \frac{5\,x^2K^2}{6\,(x^2+y^2-1)|x^2+y^2-1|} - \frac{5\,y^2K^2}{6\,(x^2+y^2-1)|x^2+y^2-1|} + \frac{x^2}{3\,(x^2+y^2-1)} \\
& + \frac{y^2}{3\,(x^2+y^2-1)} - \frac{K^2}{2\,(x^2+y^2-1)} - \frac{5\,x^2K}{3\,(x^2+y^2-1)|x^2+y^2-1|} - \frac{5\,y^2K}{3\,(x^2+y^2-1)|x^2+y^2-1|} - \frac{K}{x^2+y^2-1} - \frac{5\,x^2}{6\,(x^2+y^2-1)|x^2+y^2-1|} \\
& - \frac{5\,y^2}{6\,(x^2+y^2-1)|x^2+y^2-1|} + \frac{K^2}{2\,(x^2+y^2-1)|x^2+y^2-1|} - \frac{1}{2\,(x^2+y^2-1)} + \frac{K}{(x^2+y^2-1)|x^2+y^2-1|} + \frac{1}{2\,(x^2+y^2-1)|x^2+y^2-1|}
\end{aligned}
$$

$$
\begin{aligned}
NP18 = {} & -\frac{i\,x^6K^2\sqrt{|x^2+y^2-1|}\frac{\partial^2}{\partial x\partial y}b_x\,(u,x,y)}{8\,(x^6|K+1|+3\,x^4y^2|K+1|+3\,x^2y^4|K+1|+y^6|K+1|-3\,x^4|K+1|-6\,x^2y^2|K+1|-3\,y^4|K+1|+3\,x^2|K+1|+3\,y^2|K+1|-|K+1|)} \\
& - \frac{3i\,x^4y^2K^2\sqrt{|x^2+y^2-1|}\frac{\partial^2}{\partial x\partial y}b_x\,(u,x,y)}{8\,(x^6|K+1|+3\,x^4y^2|K+1|+3\,x^2y^4|K+1|+y^6|K+1|-3\,x^4|K+1|-6\,x^2y^2|K+1|-3\,y^4|K+1|+3\,x^2|K+1|+3\,y^2|K+1|-|K+1|)} \\
& - \frac{3i\,x^2y^4K^2\sqrt{|x^2+y^2-1|}\frac{\partial^2}{\partial x\partial y}b_x\,(u,x,y)}{8\,(x^6|K+1|+3\,x^4y^2|K+1|+3\,x^2y^4|K+1|+y^6|K+1|-3\,x^4|K+1|-6\,x^2y^2|K+1|-3\,y^4|K+1|+3\,x^2|K+1|+3\,y^2|K+1|-|K+1|)} \\
& - \frac{i\,y^6K^2\sqrt{|x^2+y^2-1|}\frac{\partial^2}{\partial x\partial y}b_x\,(u,x,y)}{8\,(x^6|K+1|+3\,x^4y^2|K+1|+3\,x^2y^4|K+1|+y^6|K+1|-3\,x^4|K+1|-6\,x^2y^2|K+1|-3\,y^4|K+1|+3\,x^2|K+1|+3\,y^2|K+1|-|K+1|)} \\
& - \frac{x^6K^2\sqrt{|x^2+y^2-1|}\frac{\partial^2}{(\partial y)^2}b_x\,(u,x,y)}{8\,(x^6|K+1|+3\,x^4y^2|K+1|+3\,x^2y^4|K+1|+y^6|K+1|-3\,x^4|K+1|-6\,x^2y^2|K+1|-3\,y^4|K+1|+3\,x^2|K+1|+3\,y^2|K+1|-|K+1|)} \\
& - \frac{3\,x^4y^2K^2\sqrt{|x^2+y^2-1|}\frac{\partial^2}{(\partial y)^2}b_x\,(u,x,y)}{8\,(x^6|K+1|+3\,x^4y^2|K+1|+3\,x^2y^4|K+1|+y^6|K+1|-3\,x^4|K+1|-6\,x^2y^2|K+1|-3\,y^4|K+1|+3\,x^2|K+1|+3\,y^2|K+1|-|K+1|)} \\
& - \frac{3\,x^2y^4K^2\sqrt{|x^2+y^2-1|}\frac{\partial^2}{(\partial y)^2}b_x\,(u,x,y)}{8\,(x^6|K+1|+3\,x^4y^2|K+1|+3\,x^2y^4|K+1|+y^6|K+1|-3\,x^4|K+1|-6\,x^2y^2|K+1|-3\,y^4|K+1|+3\,x^2|K+1|+3\,y^2|K+1|-|K+1|)} \\
& - \frac{y^6K^2\sqrt{|x^2+y^2-1|}\frac{\partial^2}{(\partial y)^2}b_x\,(u,x,y)}{8\,(x^6|K+1|+3\,x^4y^2|K+1|+3\,x^2y^4|K+1|+y^6|K+1|-3\,x^4|K+1|-6\,x^2y^2|K+1|-3\,y^4|K+1|+3\,x^2|K+1|+3\,y^2|K+1|-|K+1|)} \\
& + \frac{i\,x^6K^2\sqrt{|x^2+y^2-1|}\frac{\partial^2}{(\partial x)^2}b_y\,(u,x,y)}{8\,(x^6|K+1|+3\,x^4y^2|K+1|+3\,x^2y^4|K+1|+y^6|K+1|-3\,x^4|K+1|-6\,x^2y^2|K+1|-3\,y^4|K+1|+3\,x^2|K+1|+3\,y^2|K+1|-|K+1|)} + \dots
\end{aligned}
$$

Bianchi Identities

$$
\begin{aligned}
BI4 = {} & \frac{x^{12}K^4\sqrt{|x^2+y^2-1|}\frac{\partial^3}{(\partial x)^3}H}{8\,\mathcal{D}_6} + \frac{3\,x^{10}y^2K^4\sqrt{|x^2+y^2-1|}\frac{\partial^3}{(\partial x)^3}H}{4\,\mathcal{D}_6} + \frac{15\,x^8y^4K^4\sqrt{|x^2+y^2-1|}\frac{\partial^3}{(\partial x)^3}H}{8\,\mathcal{D}_6} \\
& + \frac{5\,x^6y^6K^4\sqrt{|x^2+y^2-1|}\frac{\partial^3}{(\partial x)^3}H}{2\,\mathcal{D}_6} + \frac{15\,x^4y^8K^4\sqrt{|x^2+y^2-1|}\frac{\partial^3}{(\partial x)^3}H}{8\,\mathcal{D}_6} + \dots
\end{aligned}
$$

$$
\begin{aligned}
\mathcal{D}_6 = {} & x^{10}|K+1| + 5\,x^8y^2|K+1| + 10\,x^6y^4|K+1| + 10\,x^4y^6|K+1| + 5\,x^2y^8|K+1| + y^{10}|K+1| - 5\,x^8|K+1| - 20\,x^6y^2|K+1| - 30\,x^4y^4|K+1| \\
& - 20\,x^2y^6|K+1| - 5\,y^8|K+1| + 10\,x^6|K+1| + 30\,x^4y^2|K+1| + 30\,x^2y^4|K+1| + 10\,y^6|K+1| - 10\,x^4|K+1| - 20\,x^2y^2|K+1| - 10\,y^4|K+1| \\
& + 5\,x^2|K+1| + 5\,y^2|K+1| - |K+1|.
\end{aligned}
$$

$$
\begin{aligned}
BI6 = {} & \frac{x^6K^2\frac{\partial}{\partial x}b_x(u,x,y)}{(x^4+2\,x^2y^2+y^4-2\,x^2-2\,y^2+1)|x^2+y^2-1|} + \frac{3\,x^4y^2K^2\frac{\partial}{\partial x}b_x(u,x,y)}{(x^4+2\,x^2y^2+y^4-2\,x^2-2\,y^2+1)|x^2+y^2-1|} + \frac{3\,x^2y^4K^2\frac{\partial}{\partial x}b_x(u,x,y)}{(x^4+2\,x^2y^2+y^4-2\,x^2-2\,y^2+1)|x^2+y^2-1|} \\
& + \frac{y^6K^2\frac{\partial}{\partial x}b_x(u,x,y)}{(x^4+2\,x^2y^2+y^4-2\,x^2-2\,y^2+1)|x^2+y^2-1|} + \frac{3i\,x^6K^2\frac{\partial}{\partial y}b_x(u,x,y)}{4\,(x^4+2\,x^2y^2+y^4-2\,x^2-2\,y^2+1)|x^2+y^2-1|} + \frac{9i\,x^4y^2K^2\frac{\partial}{\partial y}b_x(u,x,y)}{4\,(x^4+2\,x^2y^2+y^4-2\,x^2-2\,y^2+1)|x^2+y^2-1|} \\
& + \frac{9i\,x^2y^4K^2\frac{\partial}{\partial y}b_x(u,x,y)}{4\,(x^4+2\,x^2y^2+y^4-2\,x^2-2\,y^2+1)|x^2+y^2-1|} + \frac{3i\,y^6K^2\frac{\partial}{\partial y}b_x(u,x,y)}{4\,(x^4+2\,x^2y^2+y^4-2\,x^2-2\,y^2+1)|x^2+y^2-1|} - \frac{3i\,x^6K^2\frac{\partial}{\partial x}b_y(u,x,y)}{4\,(x^4+2\,x^2y^2+y^4-2\,x^2-2\,y^2+1)|x^2+y^2-1|} \\
& - \frac{9i\,x^4y^2K^2\frac{\partial}{\partial x}b_y(u,x,y)}{4\,(x^4+2\,x^2y^2+y^4-2\,x^2-2\,y^2+1)|x^2+y^2-1|} - \frac{9i\,x^2y^4K^2\frac{\partial}{\partial x}b_y(u,x,y)}{4\,(x^4+2\,x^2y^2+y^4-2\,x^2-2\,y^2+1)|x^2+y^2-1|} - \frac{3i\,y^6K^2\frac{\partial}{\partial x}b_y(u,x,y)}{4\,(x^4+2\,x^2y^2+y^4-2\,x^2-2\,y^2+1)|x^2+y^2-1|} \\
& + \frac{x^6K^2\frac{\partial}{\partial y}b_y(u,x,y)}{(x^4+2\,x^2y^2+y^4-2\,x^2-2\,y^2+1)|x^2+y^2-1|} + \frac{3\,x^4y^2K^2\frac{\partial}{\partial y}b_y(u,x,y)}{(x^4+2\,x^2y^2+y^4-2\,x^2-2\,y^2+1)|x^2+y^2-1|} + \frac{3\,x^2y^4K^2\frac{\partial}{\partial y}b_y(u,x,y)}{(x^4+2\,x^2y^2+y^4-2\,x^2-2\,y^2+1)|x^2+y^2-1|} \\
& + \frac{y^6K^2\frac{\partial}{\partial y}b_y(u,x,y)}{(x^4+2\,x^2y^2+y^4-2\,x^2-2\,y^2+1)|x^2+y^2-1|} - \frac{2\,x^5K^2b_x(u,x,y)}{(x^4+2\,x^2y^2+y^4-2\,x^2-2\,y^2+1)|x^2+y^2-1|} - \frac{3i\,x^4yK^2b_x(u,x,y)}{2\,(x^4+2\,x^2y^2+y^4-2\,x^2-2\,y^2+1)|x^2+y^2-1|} \\
& - \frac{4\,x^3y^2K^2b_x(u,x,y)}{(x^4+2\,x^2y^2+y^4-2\,x^2-2\,y^2+1)|x^2+y^2-1|} - \frac{3i\,x^2y^3K^2b_x(u,x,y)}{(x^4+2\,x^2y^2+y^4-2\,x^2-2\,y^2+1)|x^2+y^2-1|} - \frac{2\,xy^4K^2b_x(u,x,y)}{(x^4+2\,x^2y^2+y^4-2\,x^2-2\,y^2+1)|x^2+y^2-1|} \\
& - \frac{3i\,y^5K^2b_x(u,x,y)}{2\,(x^4+2\,x^2y^2+y^4-2\,x^2-2\,y^2+1)|x^2+y^2-1|} + \frac{3i\,x^5K^2b_y(u,x,y)}{2\,(x^4+2\,x^2y^2+y^4-2\,x^2-2\,y^2+1)|x^2+y^2-1|} - \frac{2\,x^4yK^2b_y(u,x,y)}{(x^4+2\,x^2y^2+y^4-2\,x^2-2\,y^2+1)|x^2+y^2-1|} \\
& + \frac{3i\,x^3y^2K^2b_y(u,x,y)}{(x^4+2\,x^2y^2+y^4-2\,x^2-2\,y^2+1)|x^2+y^2-1|} - \frac{4\,x^2y^3K^2b_y(u,x,y)}{(x^4+2\,x^2y^2+y^4-2\,x^2-2\,y^2+1)|x^2+y^2-1|} + \dots
\end{aligned}
$$

$$
\begin{aligned}
BI7 = {} & \frac{x^3K^4}{6\,(x^4|K+1| + 2\,x^2y^2|K+1| + y^4|K+1| - 2\,x^2|K+1| - 2\,y^2|K+1| + |K+1|)\sqrt{|x^2+y^2-1|}} \\
& - \frac{i\,x^2yK^4}{6\,(x^4|K+1| + 2\,x^2y^2|K+1| + y^4|K+1| - 2\,x^2|K+1| - 2\,y^2|K+1| + |K+1|)\sqrt{|x^2+y^2-1|}} \\
& + \frac{xy^2K^4}{6\,(x^4|K+1| + 2\,x^2y^2|K+1| + y^4|K+1| - 2\,x^2|K+1| - 2\,y^2|K+1| + |K+1|)\sqrt{|x^2+y^2-1|}} \\
& - \frac{i\,y^3K^4}{6\,(x^4|K+1| + 2\,x^2y^2|K+1| + y^4|K+1| - 2\,x^2|K+1| - 2\,y^2|K+1| + |K+1|)\sqrt{|x^2+y^2-1|}} \\
& + \frac{2\,x^3K^3}{3\,(x^4|K+1| + 2\,x^2y^2|K+1| + y^4|K+1| - 2\,x^2|K+1| - 2\,y^2|K+1| + |K+1|)\sqrt{|x^2+y^2-1|}} \\
& - \frac{2i\,x^2yK^3}{3\,(x^4|K+1| + 2\,x^2y^2|K+1| + y^4|K+1| - 2\,x^2|K+1| - 2\,y^2|K+1| + |K+1|)\sqrt{|x^2+y^2-1|}} \\
& + \frac{2\,xy^2K^3}{3\,(x^4|K+1| + 2\,x^2y^2|K+1| + y^4|K+1| - 2\,x^2|K+1| - 2\,y^2|K+1| + |K+1|)\sqrt{|x^2+y^2-1|}} \\
& - \frac{2i\,y^3K^3}{3\,(x^4|K+1| + 2\,x^2y^2|K+1| + y^4|K+1| - 2\,x^2|K+1| - 2\,y^2|K+1| + |K+1|)\sqrt{|x^2+y^2-1|}} \\
& + \frac{xK^4\sqrt{|x^2+y^2-1|}}{6\,(x^4|K+1| + 2\,x^2y^2|K+1| + y^4|K+1| - 2\,x^2|K+1| - 2\,y^2|K+1| + |K+1|)} - \frac{i\,yK^4\sqrt{|x^2+y^2-1|}}{6\,(x^4|K+1| + 2\,x^2y^2|K+1| + y^4|K+1| - 2\,x^2|K+1| - 2\,y^2|K+1| + |K+1|)} \\
& + \frac{x^3K^2}{(x^4|K+1| + 2\,x^2y^2|K+1| + y^4|K+1| - 2\,x^2|K+1| - 2\,y^2|K+1| + |K+1|)\sqrt{|x^2+y^2-1|}} \\
& - \frac{i\,x^2yK^2}{(x^4|K+1| + 2\,x^2y^2|K+1| + y^4|K+1| - 2\,x^2|K+1| - 2\,y^2|K+1| + |K+1|)\sqrt{|x^2+y^2-1|}} \\
& + \frac{xy^2K^2}{(x^4|K+1| + 2\,x^2y^2|K+1| + y^4|K+1| - 2\,x^2|K+1| - 2\,y^2|K+1| + |K+1|)\sqrt{|x^2+y^2-1|}} \\
& - \frac{i\,y^3K^2}{(x^4|K+1| + 2\,x^2y^2|K+1| + y^4|K+1| - 2\,x^2|K+1| - 2\,y^2|K+1| + |K+1|)\sqrt{|x^2+y^2-1|}} \\
& - \frac{xK^4}{6\,(x^4|K+1| + 2\,x^2y^2|K+1| + y^4|K+1| - 2\,x^2|K+1| - 2\,y^2|K+1| + |K+1|)\sqrt{|x^2+y^2-1|}} \\
& + \frac{i\,yK^4}{6\,(x^4|K+1| + 2\,x^2y^2|K+1| + y^4|K+1| - 2\,x^2|K+1| - 2\,y^2|K+1| + |K+1|)\sqrt{|x^2+y^2-1|}} \\
& + \frac{2\,xK^3\sqrt{|x^2+y^2-1|}}{3\,(x^4|K+1| + 2\,x^2y^2|K+1| + y^4|K+1| - 2\,x^2|K+1| - 2\,y^2|K+1| + |K+1|)} - \cdots
\end{aligned}
$$

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\begin{aligned}
BI8 = {} & \frac{x^3K^4}{6\,(x^4|K+1| + 2\,x^2y^2|K+1| + y^4|K+1| - 2\,x^2|K+1| - 2\,y^2|K+1| + |K+1|)\sqrt{|x^2+y^2-1|}} \\
& + \frac{i\,x^2yK^4}{6\,(x^4|K+1| + 2\,x^2y^2|K+1| + y^4|K+1| - 2\,x^2|K+1| - 2\,y^2|K+1| + |K+1|)\sqrt{|x^2+y^2-1|}} \\
& + \frac{xy^2K^4}{6\,(x^4|K+1| + 2\,x^2y^2|K+1| + y^4|K+1| - 2\,x^2|K+1| - 2\,y^2|K+1| + |K+1|)\sqrt{|x^2+y^2-1|}} \\
& + \frac{i\,y^3K^4}{6\,(x^4|K+1| + 2\,x^2y^2|K+1| + y^4|K+1| - 2\,x^2|K+1| - 2\,y^2|K+1| + |K+1|)\sqrt{|x^2+y^2-1|}} \\
& + \frac{2\,x^3K^3}{3\,(x^4|K+1| + 2\,x^2y^2|K+1| + y^4|K+1| - 2\,x^2|K+1| - 2\,y^2|K+1| + |K+1|)\sqrt{|x^2+y^2-1|}} \\
& + \frac{2i\,x^2yK^3}{3\,(x^4|K+1| + 2\,x^2y^2|K+1| + y^4|K+1| - 2\,x^2|K+1| - 2\,y^2|K+1| + |K+1|)\sqrt{|x^2+y^2-1|}} \\
& + \frac{2\,xy^2K^3}{3\,(x^4|K+1| + 2\,x^2y^2|K+1| + y^4|K+1| - 2\,x^2|K+1| - 2\,y^2|K+1| + |K+1|)\sqrt{|x^2+y^2-1|}} \\
& + \frac{2i\,y^3K^3}{3\,(x^4|K+1| + 2\,x^2y^2|K+1| + y^4|K+1| - 2\,x^2|K+1| - 2\,y^2|K+1| + |K+1|)\sqrt{|x^2+y^2-1|}} \\
& + \frac{xK^4\sqrt{|x^2+y^2-1|}}{6\,(x^4|K+1| + 2\,x^2y^2|K+1| + y^4|K+1| - 2\,x^2|K+1| - 2\,y^2|K+1| + |K+1|)} + \frac{i\,yK^4\sqrt{|x^2+y^2-1|}}{6\,(x^4|K+1| + 2\,x^2y^2|K+1| + y^4|K+1| - 2\,x^2|K+1| - 2\,y^2|K+1| + |K+1|)} \\
& + \frac{x^3K^2}{(x^4|K+1| + 2\,x^2y^2|K+1| + y^4|K+1| - 2\,x^2|K+1| - 2\,y^2|K+1| + |K+1|)\sqrt{|x^2+y^2-1|}} \\
& + \frac{i\,x^2yK^2}{(x^4|K+1| + 2\,x^2y^2|K+1| + y^4|K+1| - 2\,x^2|K+1| - 2\,y^2|K+1| + |K+1|)\sqrt{|x^2+y^2-1|}} \\
& + \frac{xy^2K^2}{(x^4|K+1| + 2\,x^2y^2|K+1| + y^4|K+1| - 2\,x^2|K+1| - 2\,y^2|K+1| + |K+1|)\sqrt{|x^2+y^2-1|}} \\
& + \frac{i\,y^3K^2}{(x^4|K+1| + 2\,x^2y^2|K+1| + y^4|K+1| - 2\,x^2|K+1| - 2\,y^2|K+1| + |K+1|)\sqrt{|x^2+y^2-1|}} \\
& - \frac{xK^4}{6\,(x^4|K+1| + 2\,x^2y^2|K+1| + y^4|K+1| - 2\,x^2|K+1| - 2\,y^2|K+1| + |K+1|)\sqrt{|x^2+y^2-1|}} \\
& - \frac{i\,yK^4}{6\,(x^4|K+1| + 2\,x^2y^2|K+1| + y^4|K+1| - 2\,x^2|K+1| - 2\,y^2|K+1| + |K+1|)\sqrt{|x^2+y^2-1|}} \\
& + \frac{2\,xK^3\sqrt{|x^2+y^2-1|}}{3\,(x^4|K+1| + 2\,x^2y^2|K+1| + y^4|K+1| - 2\,x^2|K+1| - 2\,y^2|K+1| + |K+1|)} + \cdots
\end{aligned}
$$

$$
\begin{aligned}
BI10 = &-\frac{x^3K^4}{8\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}}\\
&-\frac{i\,x^2yK^4}{8\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}}\\
&-\frac{xy^2K^4}{8\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}}\\
&-\frac{i\,y^3K^4}{8\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}}\\
&-\frac{x^3K^3}{2\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}}\\
&-\frac{i\,x^2yK^3}{2\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}}\\
&-\frac{xy^2K^3}{2\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}}\\
&-\frac{i\,y^3K^3}{2\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}}\\
&-\frac{xK^4\sqrt{|x^2+y^2-1|}}{8\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)}-\frac{i\,yK^4\sqrt{|x^2+y^2-1|}}{8\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)}\\
&-\frac{3\,x^3K^2}{4\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}}\\
&-\frac{3i\,x^2yK^2}{4\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}}\\
&-\frac{3\,xy^2K^2}{4\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}}\\
&-\frac{3i\,y^3K^2}{4\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}}\\
&+\frac{xK^4}{8\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}}\\
&+\frac{i\,yK^4}{8\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)\sqrt{|x^2+y^2-1|}}\\
&-\frac{xK^3\sqrt{|x^2+y^2-1|}}{2\left(x^4|K+1|+2\,x^2y^2|K+1|+y^4|K+1|-2\,x^2|K+1|-2\,y^2|K+1|+|K+1|\right)}-\cdots
\end{aligned}
$$

$$
\begin{aligned}
BI11 = {} & \frac{x^4 K^2 \frac{\partial}{\partial x} b_x(u,x,y)}{(x^2+y^2-1)|x^2+y^2-1|} + \frac{2\,x^2 y^2 K^2 \frac{\partial}{\partial x} b_x(u,x,y)}{(x^2+y^2-1)|x^2+y^2-1|} + \frac{y^4 K^2 \frac{\partial}{\partial x} b_x(u,x,y)}{(x^2+y^2-1)|x^2+y^2-1|} + \frac{x^4 K^2 \frac{\partial}{\partial y} b_y(u,x,y)}{(x^2+y^2-1)|x^2+y^2-1|} + \frac{2\,x^2 y^2 K^2 \frac{\partial}{\partial y} b_y(u,x,y)}{(x^2+y^2-1)|x^2+y^2-1|} \\
& + \frac{y^4 K^2 \frac{\partial}{\partial y} b_y(u,x,y)}{(x^2+y^2-1)|x^2+y^2-1|} - \frac{2\,x^3 K^2 b_x(u,x,y)}{(x^2+y^2-1)|x^2+y^2-1|} - \frac{2\,x y^2 K^2 b_x(u,x,y)}{(x^2+y^2-1)|x^2+y^2-1|} - \frac{2\,x^2 y K^2 b_y(u,x,y)}{(x^2+y^2-1)|x^2+y^2-1|} - \frac{2\,y^3 K^2 b_y(u,x,y)}{(x^2+y^2-1)|x^2+y^2-1|} \\
& + \frac{x^4 K \frac{\partial}{\partial u} K}{2\,(x^2+y^2-1)|x^2+y^2-1|} + \frac{x^2 y^2 K \frac{\partial}{\partial u} K}{(x^2+y^2-1)|x^2+y^2-1|} + \frac{y^4 K \frac{\partial}{\partial u} K}{2\,(x^2+y^2-1)|x^2+y^2-1|} + \frac{x^2 K^2 \frac{\partial}{\partial x} b_x(u,x,y)}{x^2+y^2-1} + \frac{y^2 K^2 \frac{\partial}{\partial x} b_x(u,x,y)}{x^2+y^2-1} + \frac{2\,x^4 K \frac{\partial}{\partial x} b_x(u,x,y)}{(x^2+y^2-1)|x^2+y^2-1|} \\
& + \frac{4\,x^2 y^2 K \frac{\partial}{\partial x} b_x(u,x,y)}{(x^2+y^2-1)|x^2+y^2-1|} + \frac{2\,y^4 K \frac{\partial}{\partial x} b_x(u,x,y)}{(x^2+y^2-1)|x^2+y^2-1|} + \frac{x^2 K^2 \frac{\partial}{\partial y} b_y(u,x,y)}{x^2+y^2-1} + \frac{y^2 K^2 \frac{\partial}{\partial y} b_y(u,x,y)}{x^2+y^2-1} + \frac{2\,x^4 K \frac{\partial}{\partial y} b_y(u,x,y)}{(x^2+y^2-1)|x^2+y^2-1|} + \frac{4\,x^2 y^2 K \frac{\partial}{\partial y} b_y(u,x,y)}{(x^2+y^2-1)|x^2+y^2-1|} \\
& + \frac{2\,y^4 K \frac{\partial}{\partial y} b_y(u,x,y)}{(x^2+y^2-1)|x^2+y^2-1|} - \frac{2\,x K^2 b_x(u,x,y)}{x^2+y^2-1} - \frac{4\,x^3 K b_x(u,x,y)}{(x^2+y^2-1)|x^2+y^2-1|} - \frac{4\,x y^2 K b_x(u,x,y)}{(x^2+y^2-1)|x^2+y^2-1|} - \frac{2\,y K^2 b_y(u,x,y)}{x^2+y^2-1} - \frac{4\,x^2 y K b_y(u,x,y)}{(x^2+y^2-1)|x^2+y^2-1|} \\
& - \frac{4\,y^3 K b_y(u,x,y)}{(x^2+y^2-1)|x^2+y^2-1|} + \frac{x^2 K \frac{\partial}{\partial u} K}{2\,(x^2+y^2-1)} + \frac{y^2 K \frac{\partial}{\partial u} K}{2\,(x^2+y^2-1)} + \frac{x^4 \frac{\partial}{\partial u} K}{2\,(x^2+y^2-1)|x^2+y^2-1|} + \frac{x^2 y^2 \frac{\partial}{\partial u} K}{(x^2+y^2-1)|x^2+y^2-1|} + \frac{y^4 \frac{\partial}{\partial u} K}{2\,(x^2+y^2-1)|x^2+y^2-1|} \\
& + \frac{2\,x^2 K \frac{\partial}{\partial x} b_x(u,x,y)}{x^2+y^2-1} + \frac{2\,y^2 K \frac{\partial}{\partial x} b_x(u,x,y)}{x^2+y^2-1} + \frac{x^4 \frac{\partial}{\partial x} b_x(u,x,y)}{(x^2+y^2-1)|x^2+y^2-1|} + \frac{2\,x^2 y^2 \frac{\partial}{\partial x} b_x(u,x,y)}{(x^2+y^2-1)|x^2+y^2-1|} + \frac{y^4 \frac{\partial}{\partial x} b_x(u,x,y)}{(x^2+y^2-1)|x^2+y^2-1|} - \frac{2\,x^2 K^2 \frac{\partial}{\partial x} b_x(u,x,y)}{(x^2+y^2-1)|x^2+y^2-1|} - \cdots
\end{aligned}
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Petrov Type

Type "II"